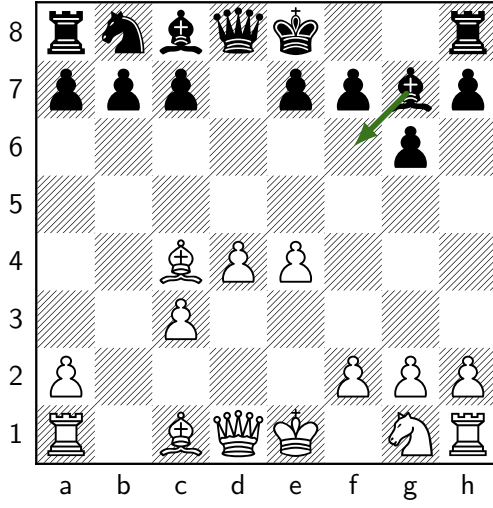


Melanie-MASTERS (human masters) — Grunfeld

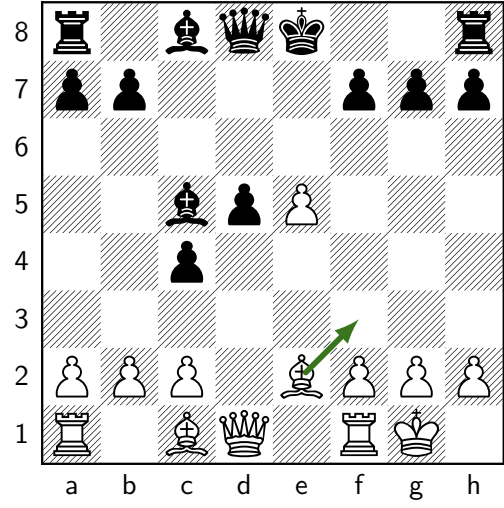
Melanie (Deep_Vision)

Abstract

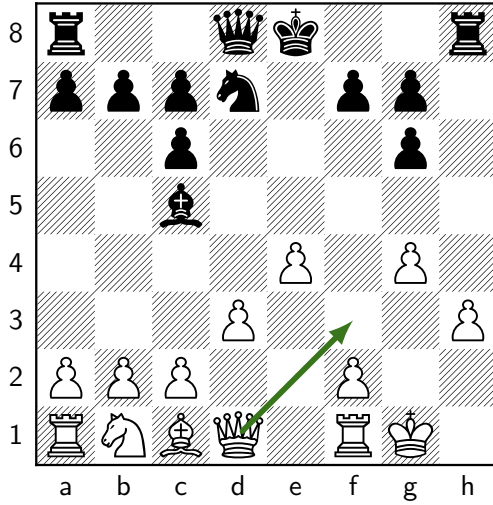
Every move below was produced by relational analogy to named master games (human and/or top-engine – the exact source is named per move). Each figure shows our position with Melanie's move (a) and the top retrieved analog positions (b–d) with the master move played there; the caption names the exact source and the piece-role correspondence under μ – cross-type maps (a rook playing a queen's role) are called out, as no material/eval engine explains a move that way.



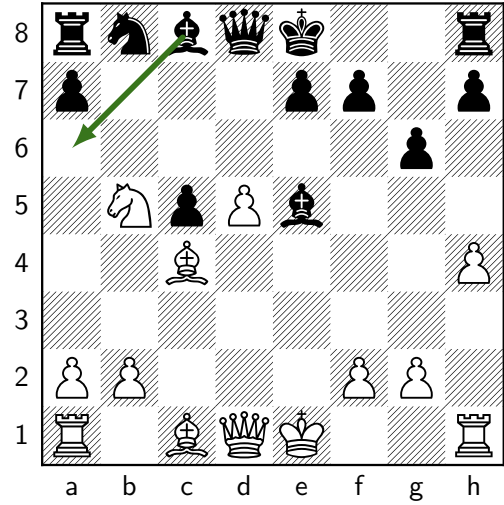
(a)



(b)

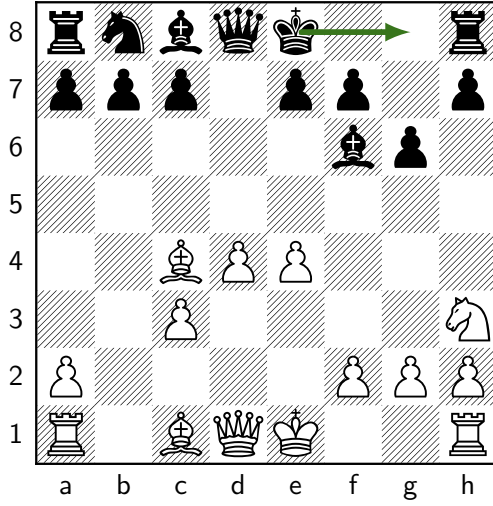


(c)

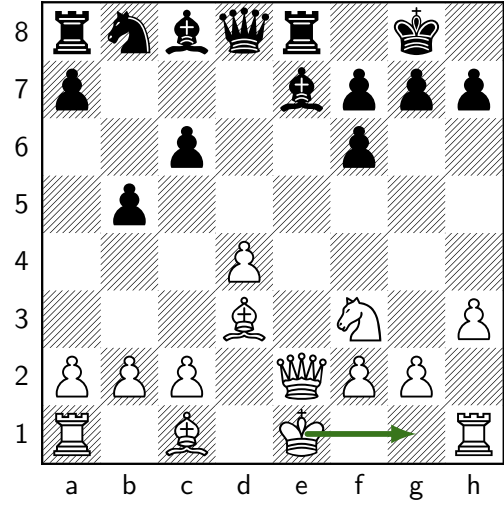


(d)

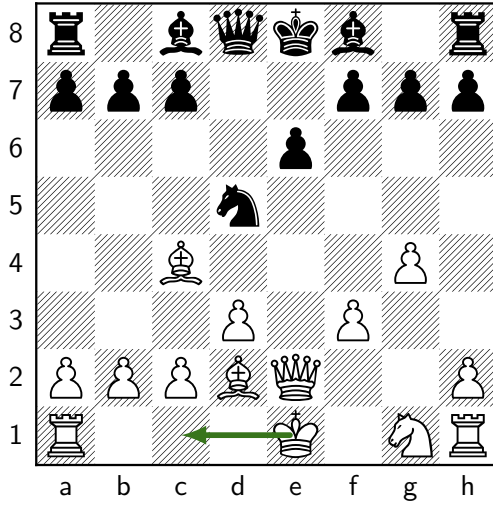
Figure 1: **Move 7, Black: Bf6.** (a) Our position; Melanie plays Bf6 (g7f6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0028#1383` (ply 22) [structural distance $d=65$, QAP obj=62]: the master played Bf3; under μ our Bg2 plays the role of that game’s Be2; cross-type roles: our Pf7 as their Ra1, our Qd8 as their bc5; transferred move g7f6 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0017#2613` (ply 20) [structural distance $d=74$, QAP obj=45]: the master played Qf3; under μ our Bg2 plays the role of that game’s Qd1 (a **cross-type** correspondence); cross-type roles: our pc3 as their Nb1, our Pg6 as their Bc1; transferred move g7f6 (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0024#907` (ply 21) [structural distance $d=62$, QAP obj=73]: the master played Ba6; under μ our Bc1 plays the role of that game’s Bc1; cross-type roles: our Bg7 as their Nb8, our bc4 as their nb5; transferred move c8f5. Synthesis: 2 retrieved analogs transferred to Bf6 (summed QAP objective 107, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



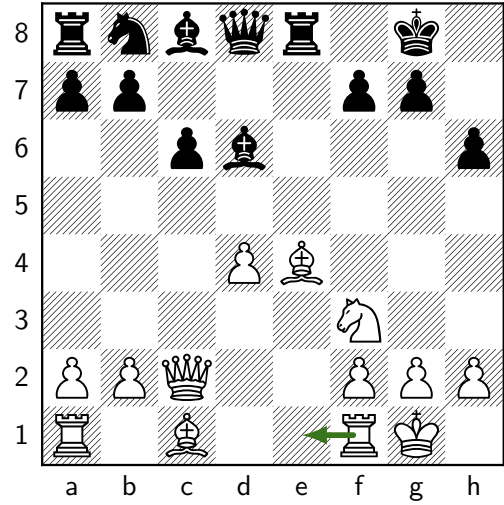
(a)



(b)

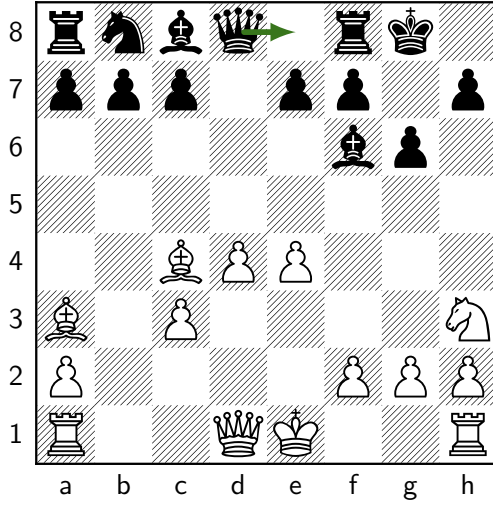


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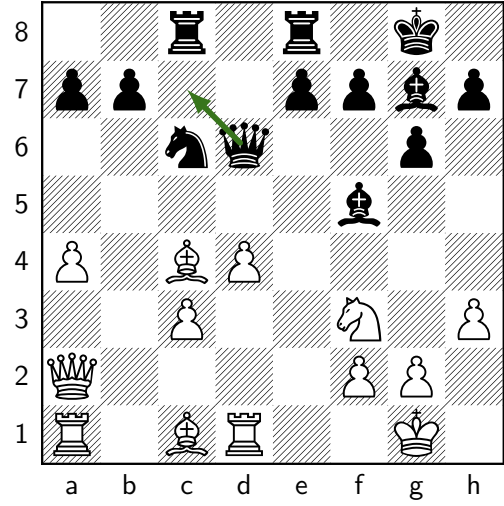


(d)

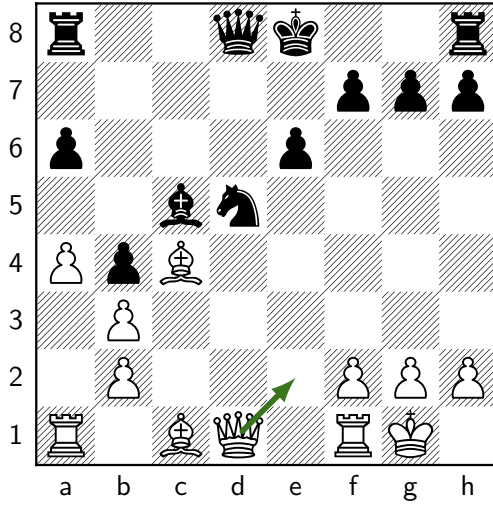
Figure 2: **Move 8, Black: 0-0.** (a) Our position; Melanie plays 0-0 (e8g8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0028#1338` (ply 20) [structural distance $d=59$, QAP obj=60]: the master played 0-0; under μ our *Ke1* plays the role of that game’s *Ke1*; cross-type roles: our *Nb8* as their *Ra1*, our *pe4* as their *Bc1*; transferred move e8g8 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0026#1106` (ply 22) [structural distance $d=72$, QAP obj=34]: the master played 0-0-0; under μ our *Ke1* plays the role of that game’s *Ke1*; cross-type roles: our *Bc8* as their *Ra1*, our *Pg6* as their *Ng1*; transferred move e8g8 (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0019#1492` (ply 24) [structural distance $d=72$, QAP obj=70]: the master played *Re1*; under μ our *Rh1* plays the role of that game’s *Rf1*; cross-type roles: our *Nb8* as their *Pa2*, our *pf2* as their *Nf3*; transferred move h8g8. Synthesis: 2 retrieved analogs transferred to 0-0 (summed QAP objective 94, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



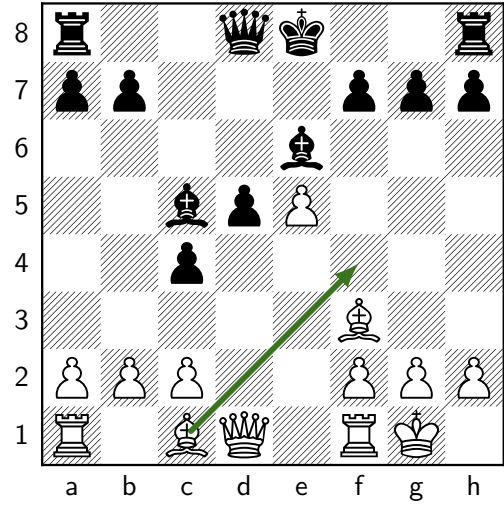
(a)



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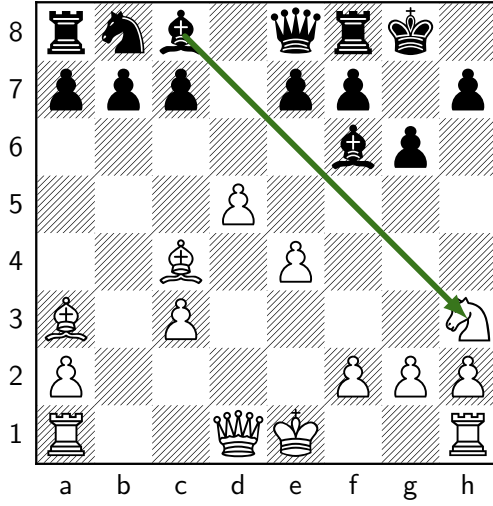


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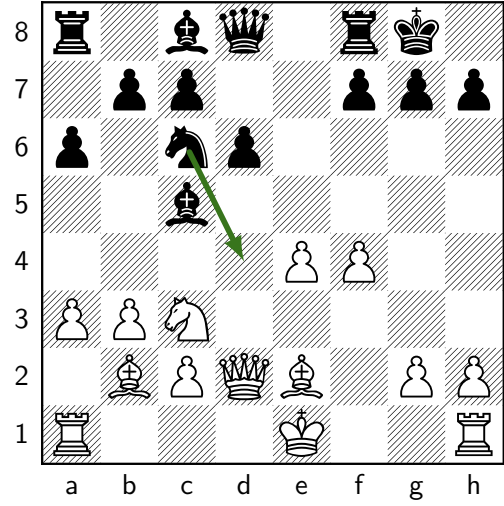


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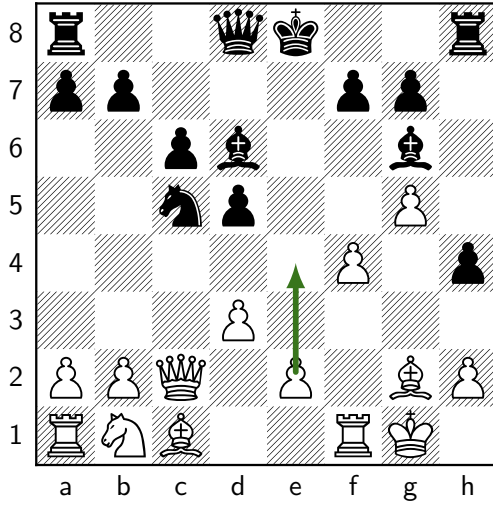
Figure 3: **Move 9, Black: Qe8**. (a) Our position; Melanie plays Qe8 (d8e8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0003#3615` (ply 31) [structural distance $d=78$, QAP obj=103]: the master played Qc7; under μ our Qd1 plays the role of that game’s Qd3; cross-type roles: our Pb7 as their Rc8, our Pe7 as their Re8; transferred move d8e8 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0025#695` (ply 28) [structural distance $d=76$, QAP obj=45]: the master played Qe2; under μ our Qd1 plays the role of that game’s Qd1; cross-type roles: our Pg6 as their Ra1, our Ra8 as their Ph2; transferred move d8e8 (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0028#1383` (ply 24) [structural distance $d=81$, QAP obj=107]: the master played Bf4; under μ our Bc1 plays the role of that game’s Bc1; cross-type roles: our Nb8 as their pb7, our nh3 as their pf7; transferred move c8e6. Synthesis: 2 retrieved analogs transferred to Qe8 (summed QAP objective 148, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



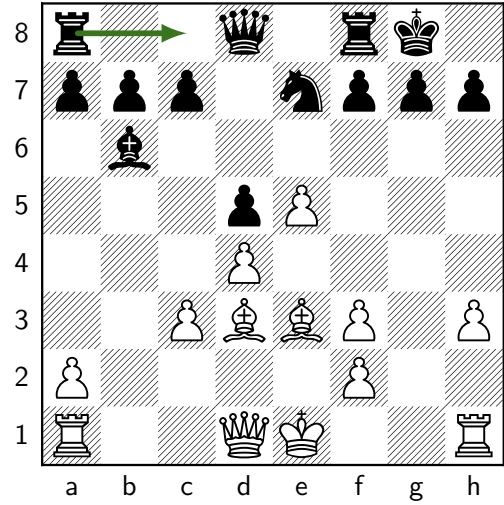
(a)



(b)

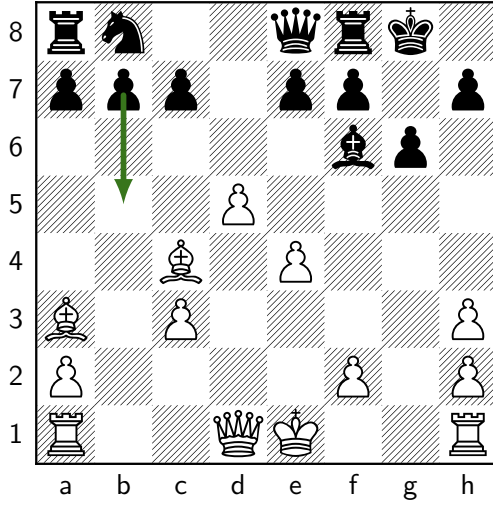


(c)

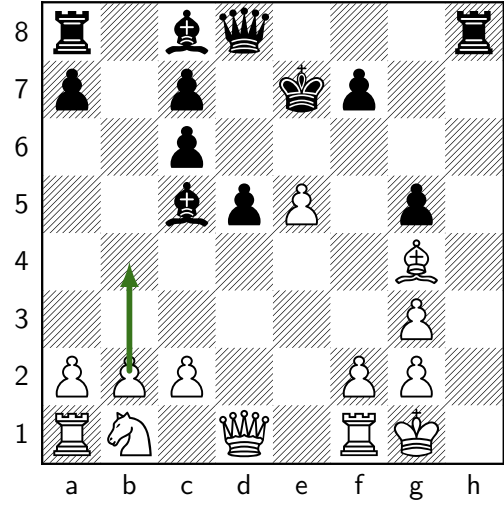


(d)

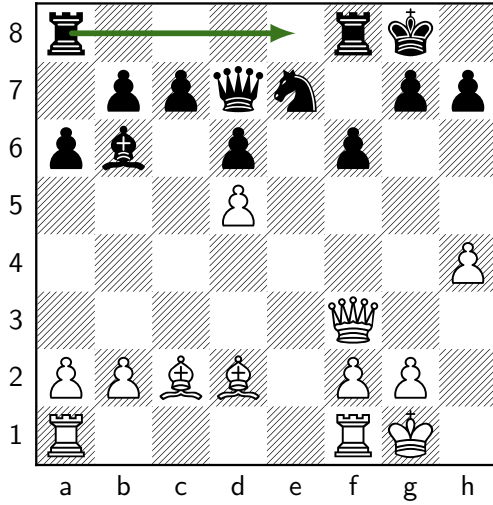
Figure 4: **Move 10, Black: Bxh3**. (a) Our position; Melanie plays Bxh3 (c8h3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0024#2197` (ply 27) [structural distance $d=67$, QAP obj=51]: the master played Nd4; under μ our *Pb2* plays the role of that game’s *Nc3* (a **cross-type** correspondence); cross-type roles: our *Pf7* as their *Bc8*, our *pa2* as their *Qd8*; transferred move b7b5. (c) Retrieved from `lichess_elite_2024-09_p0020#2744` (ply 28) [structural distance $d=60$, QAP obj=46]: the master played e4; under μ our *Pb2* plays the role of that game’s *Pe2*; cross-type roles: our *Nb8* as their *Ra1*, our *Rf8* as their *Nb1*; transferred move b7b5. (d) Retrieved from `lichess_elite_2024-09_p0005#2498` (ply 27) [structural distance $d=63$, QAP obj=41]: the master played Rc8; under μ our *Pc2* plays the role of that game’s *Ra1* (a **cross-type** correspondence); cross-type roles: our *ra1* as their *Pc7*, our *bc4* as their *Ne7*; transferred move c7c6. Synthesis: no retrieved analog transferred to Bxh3 – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



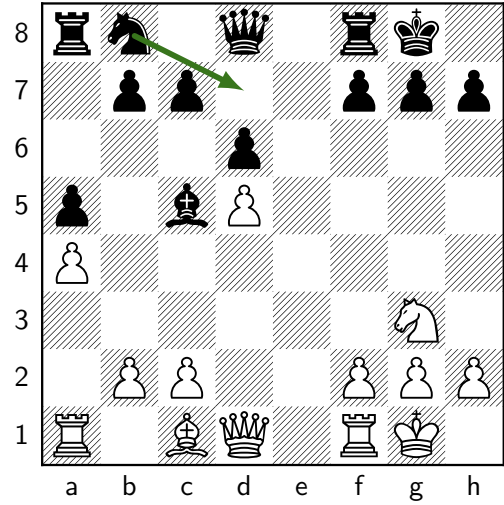
(a)



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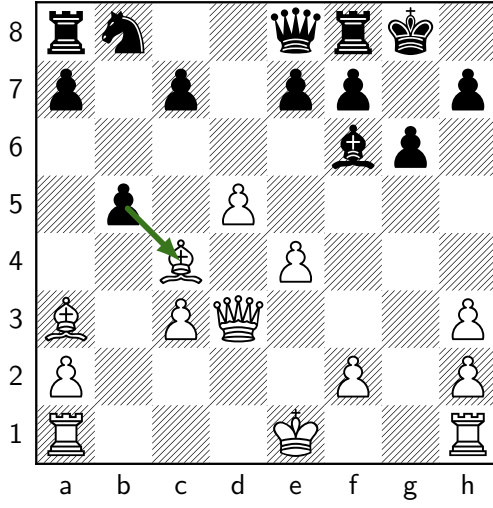


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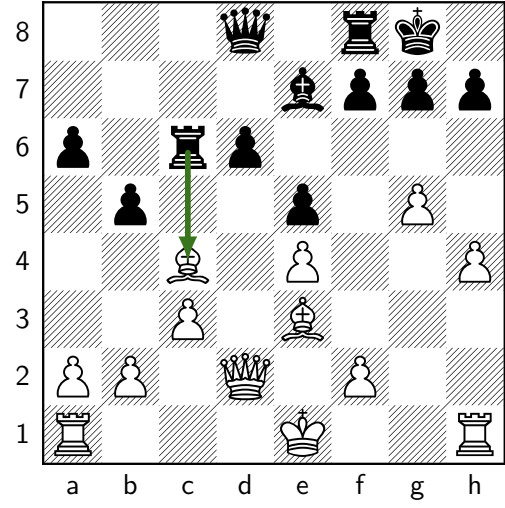


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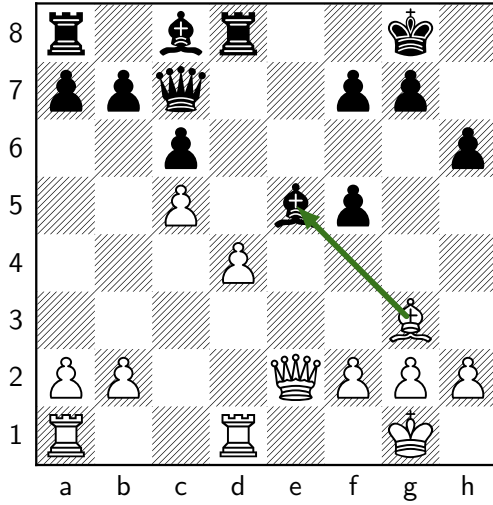
Figure 5: **Move 11, Black: b5.** (a) Our position; Melanie plays b5 (b7b5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0026#3181` (ply 26) [structural distance $d=64$, QAP obj=54]: the master played b4; under μ our *Pb2* plays the role of that game’s *Pb2*; cross-type roles: our *Qe8* as their *Nb1*, our *Bf6* as their *Qd1*; transferred move b7b5 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0003#2024` (ply 33) [structural distance $d=72$, QAP obj=63]: the master played *Rae8*; under μ our *Nb1* plays the role of that game’s *Ra1* (a **cross-type** correspondence); cross-type roles: our *bc4* as their *Qd7*, our *ra1* as their *Ne7*; transferred move b8d7. (d) Retrieved from `lichess_elite_2024-09_p0026#1803` (ply 23) [structural distance $d=64$, QAP obj=53]: the master played *Nd7*; under μ our *Nb1* plays the role of that game’s *Nb1*; cross-type roles: our *Pg6* as their *Qd8*, our *Qe8* as their *Pb7*; transferred move b8d7. Synthesis: 1 retrieved analog transferred to b5 (summed QAP objective 54, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



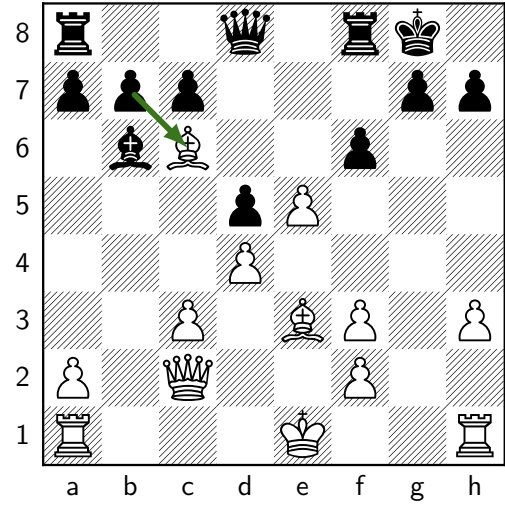
(a)



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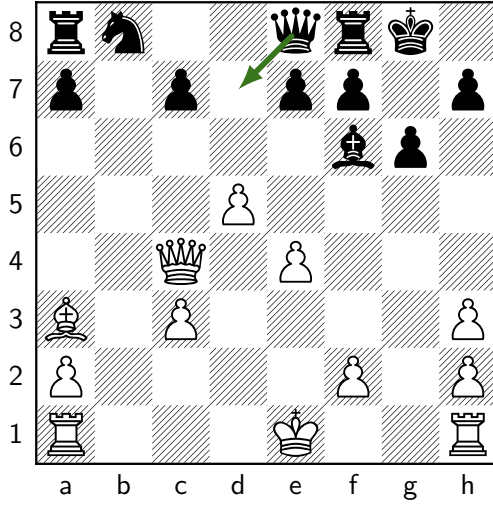


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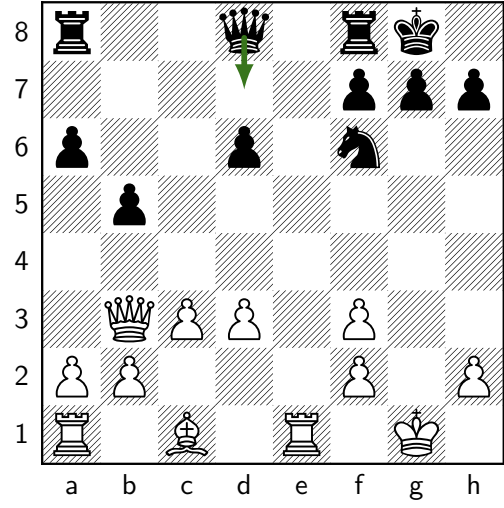


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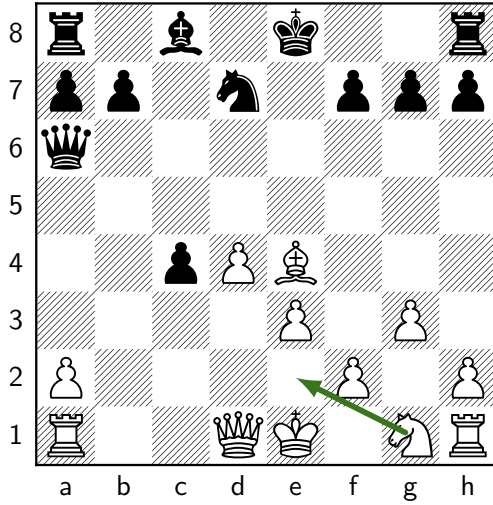
Figure 6: **Move 12, Black: bxc4**. (a) Our position; Melanie plays **bxc4** (**b5c4**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0010#4000` (ply 35) [structural distance $d=82$, QAP obj=57]: the master played **Rxc4**; under μ our Pb_4 plays the role of that game’s $Rc3$ (a **cross-type** correspondence); cross-type roles: our $Ra8$ as their $Pa6$, our $Nb8$ as their $pe4$; transferred move **b5c4** (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0030#2192` (ply 34) [structural distance $d=81$, QAP obj=49]: the master played **Bxe5**; under μ our Pb_4 plays the role of that game’s $Bg3$ (a **cross-type** correspondence); cross-type roles: our $Qe8$ as their $Rd1$, our $Nb8$ as their $Pa2$; transferred move **b5c4** (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0010#3095` (ply 29) [structural distance $d=78$, QAP obj=45]: the master played **bxc6**; under μ our Pb_4 plays the role of that game’s $Pb2$; cross-type roles: our $ba3$ as their $Pg7$, our $bc4$ as their $Pf6$; transferred move **b5c4** (**endorses this move**). Synthesis: 4 retrieved analogs transferred to **bxc4** (summed QAP objective 181, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



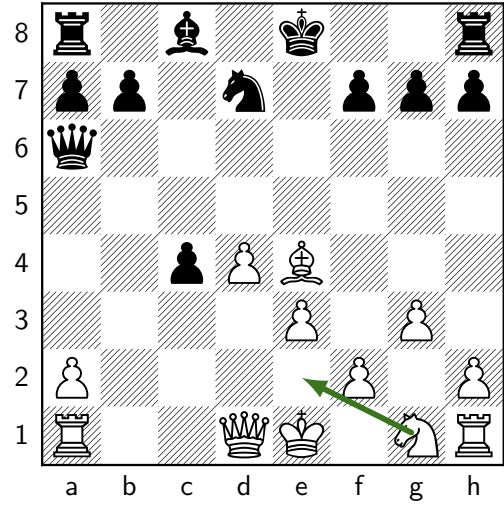
(a)



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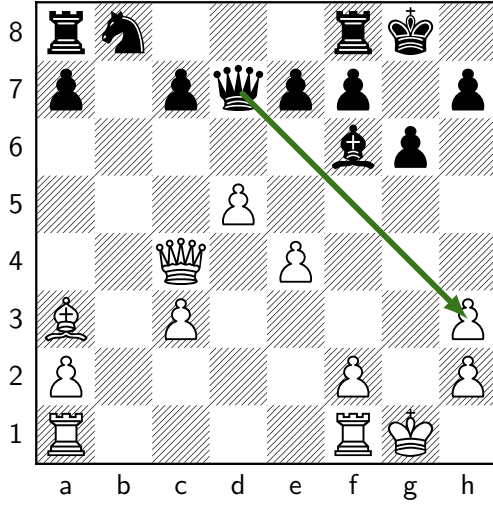


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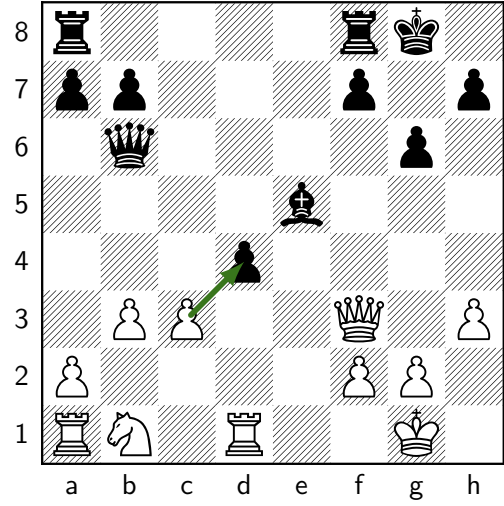


(d)

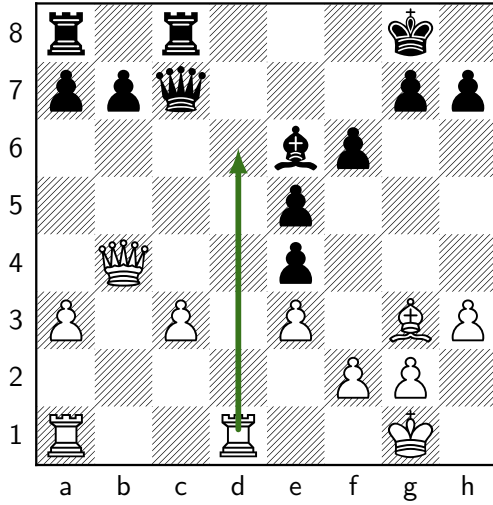
Figure 7: **Move 13, Black: Qd7.** (a) Our position; Melanie plays Qd7 (e8d7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0017#2301` (ply 31) [structural distance $d=66$, QAP obj=42]: the master played Qd7; under μ our Qe1 plays the role of that game’s Qd1; cross-type roles: our pe4 as their Ra8, our Ra8 as their Ph7; transferred move e8d7 (**endorses this move**). (c) Retrieved from `lichess_elite_2025-11_p0000#2007` (ply 24) [structural distance $d=70$, QAP obj=90]: the master played Ne2; under μ our Pg3 plays the role of that game’s Ng1 (a **cross-type** correspondence); cross-type roles: our pd5 as their Ra1, our Rf8 as their Qd1; transferred move g6g5. (d) Retrieved from `lichess_elite_2025-11_p0000#2081` (ply 24) [structural distance $d=70$, QAP obj=90]: the master played Ne2; under μ our Pg3 plays the role of that game’s Ng1 (a **cross-type** correspondence); cross-type roles: our pd5 as their Ra1, our Rf8 as their Qd1; transferred move g6g5. Synthesis: 1 retrieved analog transferred to Qd7 (summed QAP objective 42, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



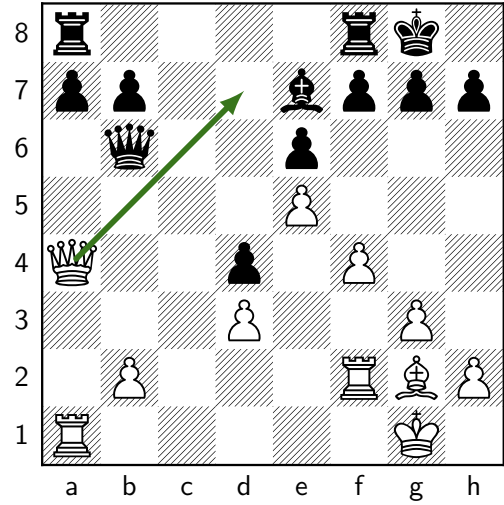
(a)



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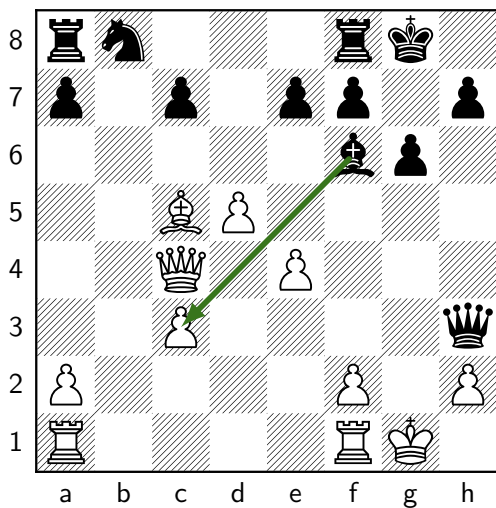


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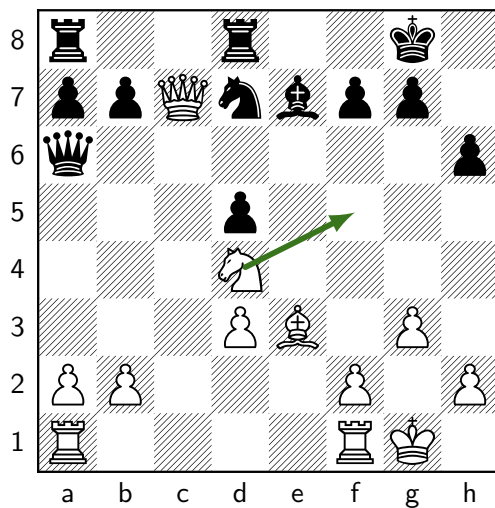


(d)

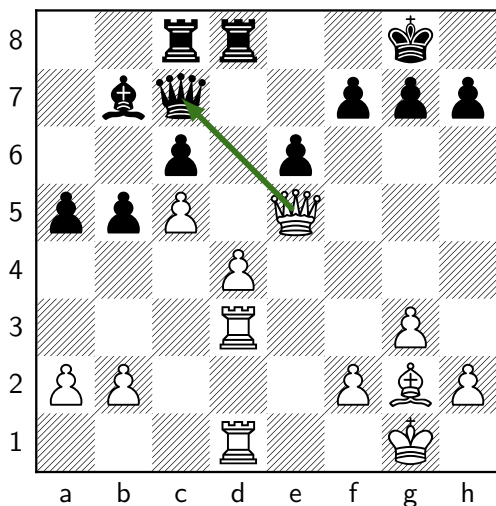
Figure 8: **Move 14, Black: Qxh3.** (a) Our position; Melanie plays Qxh3 (d7h3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0018#1371` (ply 36) [structural distance $d=82$, QAP obj=80]: the master played cxd4; under μ our Qd2 plays the role of that game’s Pc3 (a **cross-type** correspondence); cross-type roles: our pe4 as their Ra1, our Qd7 as their Rd1; transferred move d7d5. (c) Retrieved from `lichess_elite_2024-09_p0030#1871` (ply 42) [structural distance $d=63$, QAP obj=80]: the master played Rd6; under μ our Nb1 plays the role of that game’s Rd1 (a **cross-type** correspondence); cross-type roles: our Rf8 as their ph7, our Pa7 as their ra8; transferred move b8a6. (d) Retrieved from `lichess_elite_2024-09_p0010#1931` (ply 42) [structural distance $d=82$, QAP obj=75]: the master played Qd7; under μ our Bf3 plays the role of that game’s Qa4 (a **cross-type** correspondence); cross-type roles: our Pa7 as their Ra1, our Nb8 as their Pd3; transferred move f6h4. Synthesis: no retrieved analog transferred to Qxh3 – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



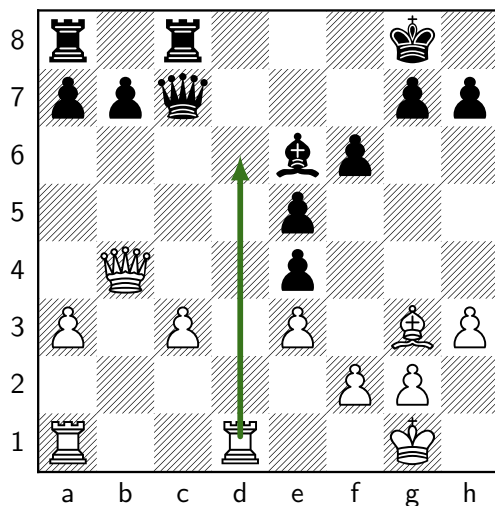
(a)



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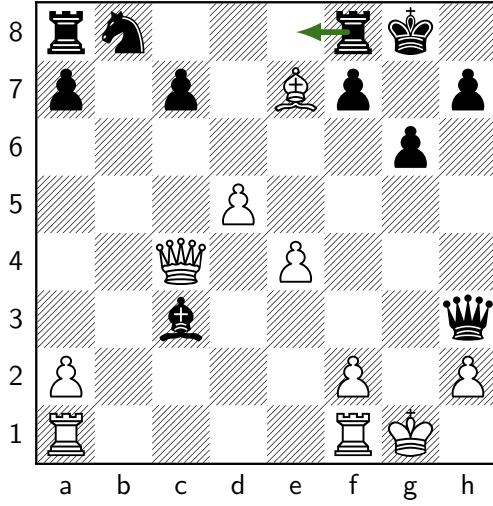


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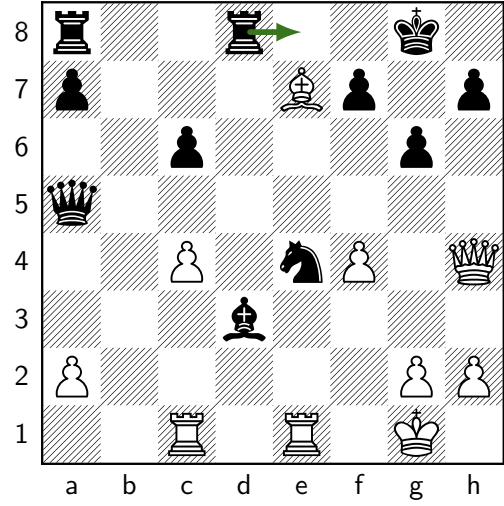


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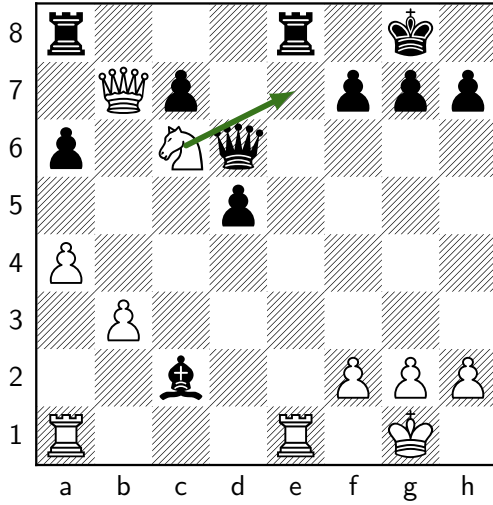
Figure 9: **Move 15, Black: Bxc3**. (a) Our position; Melanie plays Bxc3 (f6c3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0021#2927` (ply 34) [structural distance $d=80$, QAP obj=51]: the master played Nf5; under μ our *Pc2* plays the role of that game’s *Nd4* (a **cross-type** correspondence); cross-type roles: our *Bf6* as their *Ra1*, our *Ra8* as their *Pb2*; transferred move c7c6. (c) Retrieved from `lichess_elite_2025-11_p0000#2130` (ply 42) [structural distance $d=67$, QAP obj=40]: the master played Qxc7; under μ our *Qh6* plays the role of that game’s *Qe5*; cross-type roles: our *Ra8* as their *pa5*, our *Nb8* as their *pb5*; transferred move h3d3. (d) Retrieved from `lichess_elite_2024-09_p0030#1871` (ply 42) [structural distance $d=62$, QAP obj=26]: the master played Rd6; under μ our *Rf1* plays the role of that game’s *Rd1*; cross-type roles: our *qc4* as their *Pg2*, our *Bf6* as their *Qb4*; transferred move f8d8. Synthesis: no retrieved analog transferred to Bxc3 – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



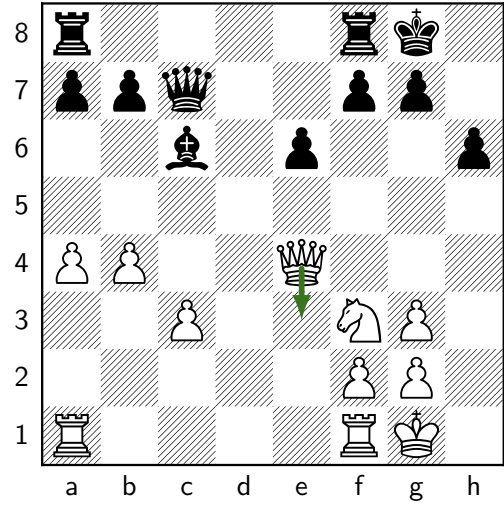
(a)



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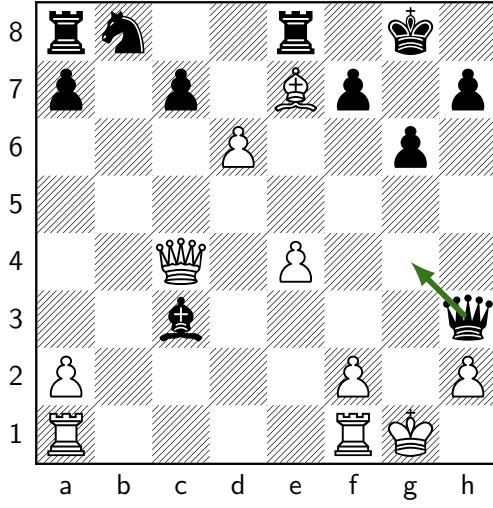


(c)

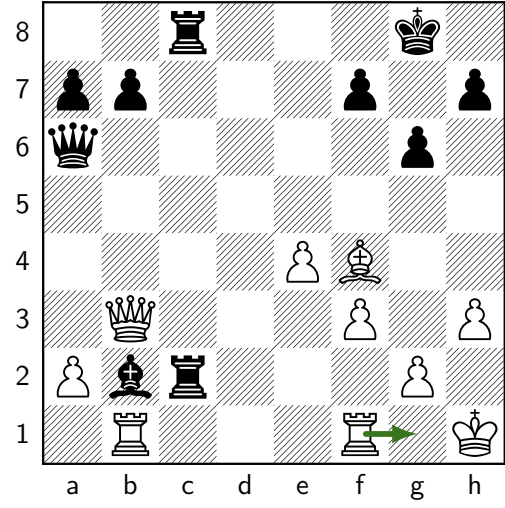


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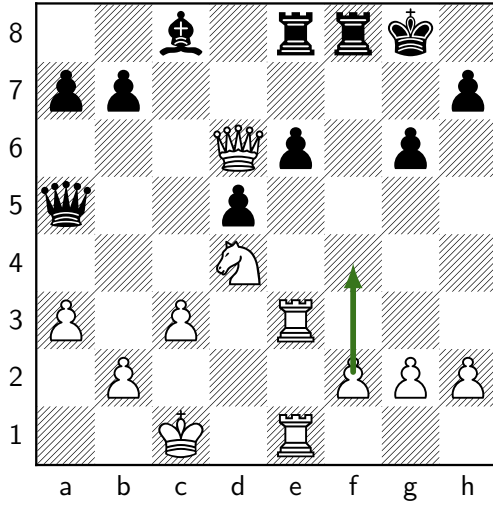
Figure 10: **Move 16, Black: Re8.** (a) Our position; Melanie plays Re8 (f8e8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0025#156` (ply 47) [structural distance $d=66$, QAP obj=36]: the master played Re8; under μ our Rf1 plays the role of that game’s Rd1; cross-type roles: our Nb8 as their Ra8, our ra1 as their Ne4; transferred move f8e8 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0025#3259` (ply 46) [structural distance $d=70$, QAP obj=31]: the master played Ne7+; under μ our Bc6 plays the role of that game’s Nc6 (a **cross-type** correspondence); cross-type roles: our Ph7 as their Ra1, our pf2 as their Re1; transferred move c3e1. (d) Retrieved from `lichess_elite_2024-09_p0020#3229` (ply 36) [structural distance $d=69$, QAP obj=19]: the master played Qe3; under μ our Rf1 plays the role of that game’s Qe4 (a **cross-type** correspondence); cross-type roles: our Bc3 as their Ra1, our Pf7 as their Rf1; transferred move f8c8. Synthesis: 1 retrieved analog transferred to Re8 (summed QAP objective 36, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



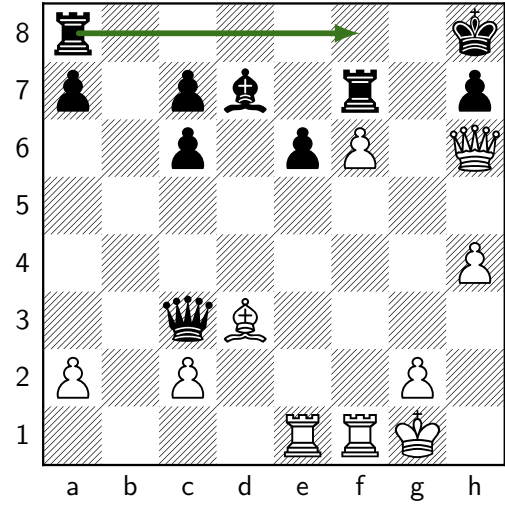
(a)



(b)

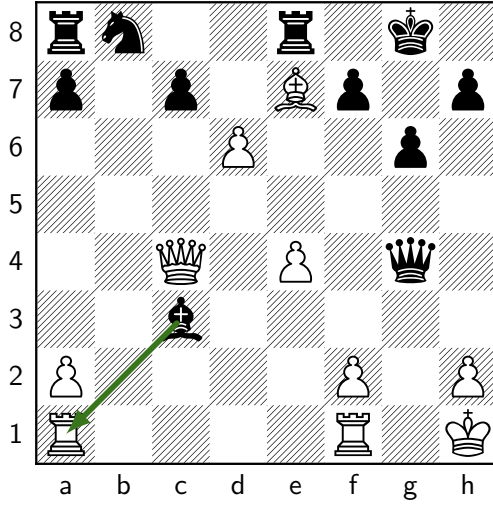


(c)

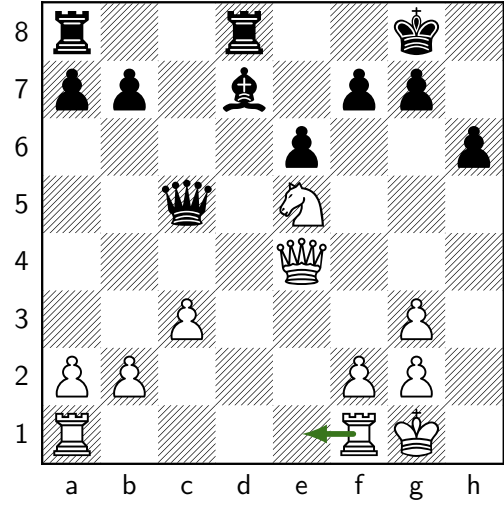


(d)

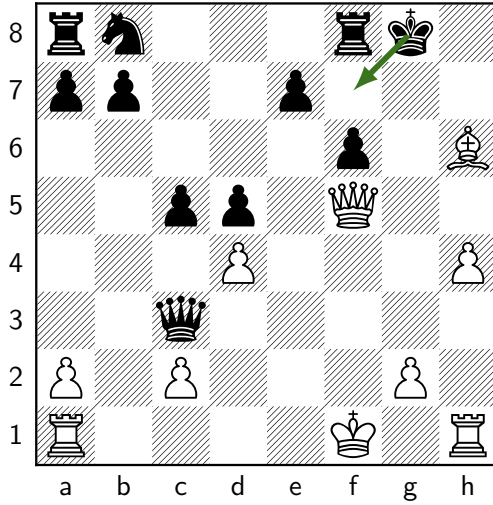
Figure 11: **Move 17, Black: Qg4+.** (a) Our position; Melanie plays Qg4+ (h3g4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0013#3652` (ply 44) [structural distance $d=74$, QAP obj=62]: the master played Rg1; under μ our Re1 plays the role of that game’s Rf1; cross-type roles: our qc4 as their Rb1, our Bc3 as their rc2; transferred move e8f8. (c) Retrieved from `lichess_elite_2024-09_p0016#1970` (ply 46) [structural distance $d=75$, QAP obj=60]: the master played f4; under μ our Pc2 plays the role of that game’s Pf2; cross-type roles: our pa2 as their Re1, our Re8 as their Pb2; transferred move c7d6. (d) Retrieved from `lichess_elite_2024-09_p0013#1264` (ply 47) [structural distance $d=78$, QAP obj=58]: the master played Raf8; under μ our Nb1 plays the role of that game’s Ra1 (a **cross-type** correspondence); cross-type roles: our Ra8 as their Pa7, our Ph7 as their Bd7; transferred move b8d7. Synthesis: no retrieved analog transferred to Qg4+ – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



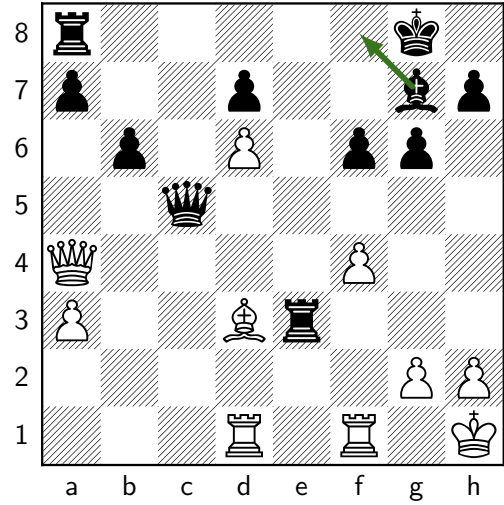
(a)



(b)

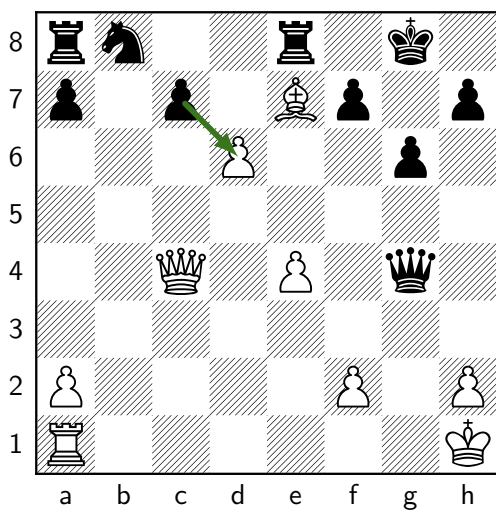


(c)

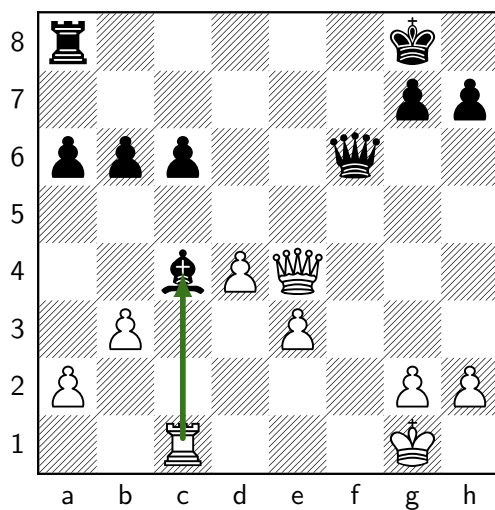


(d)

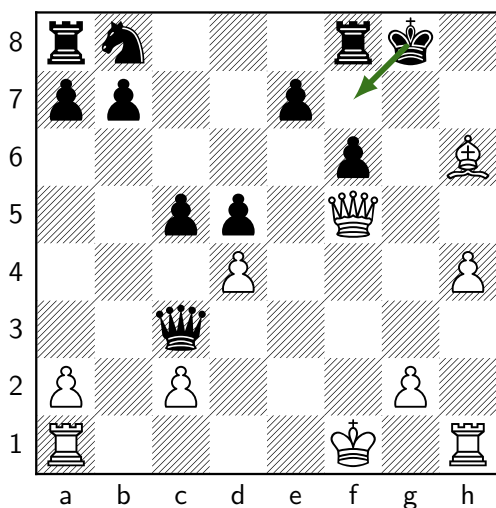
Figure 12: **Move 18, Black: Bxa1.** (a) Our position; Melanie plays Bxa1 (c3a1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0028#1041` (ply 34) [structural distance $d=78$, QAP obj=84]: the master played Rfe1; under μ our Re1 plays the role of that game’s Rf1; cross-type roles: our Nb8 as their Pa2, our Bc3 as their Ne5; transferred move e8c8. (c) Retrieved from `lichess_elite_2024-09_p0016#857` (ply 35) [structural distance $d=73$, QAP obj=65]: the master played Kf7; under μ our Kg1 plays the role of that game’s Kg1; cross-type roles: our Nb8 as their Ra8, our rf1 as their Nb8; transferred move g8g7. (d) Retrieved from `lichess_elite_2024-09_p0031#1320` (ply 47) [structural distance $d=76$, QAP obj=47]: the master played Bf8; under μ our Pg3 plays the role of that game’s Bg2 (a **cross-type** correspondence); cross-type roles: our Ph7 as their Ra8, our Bc3 as their Pd7; transferred move g6g5. Synthesis: no retrieved analog transferred to Bxa1 – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



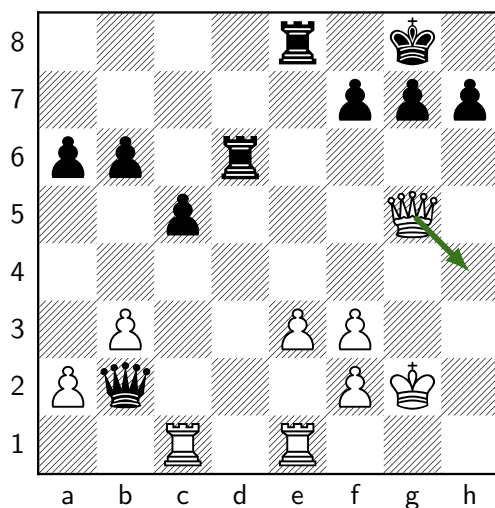
(a)



(b)

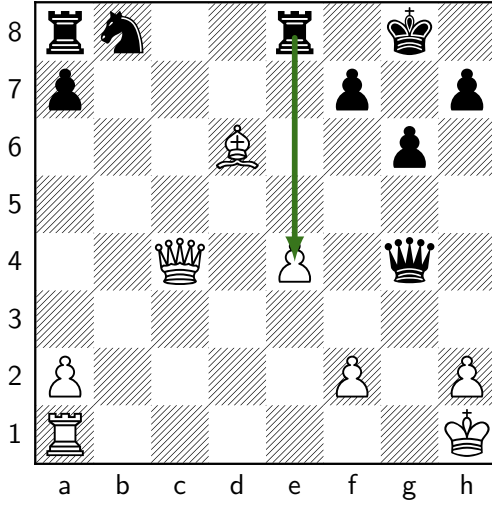


(c)

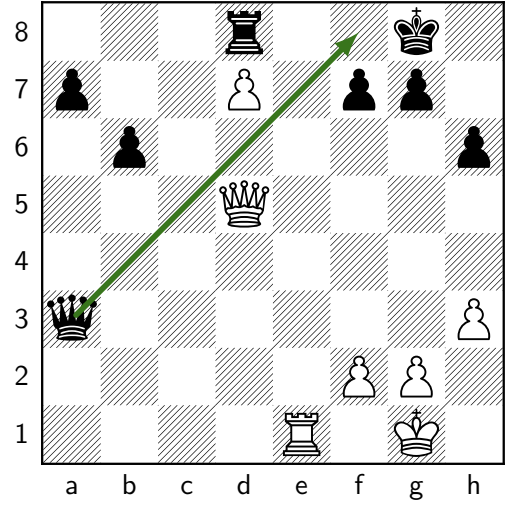


(d)

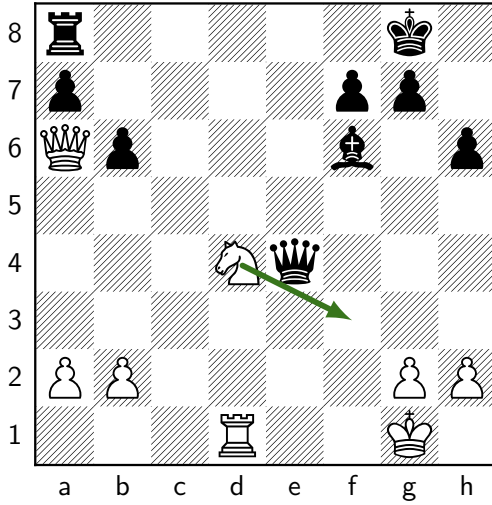
Figure 13: **Move 19, Black: cxd6.** (a) Our position; Melanie plays cxd6 (c7d6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0003#3152` (ply 42) [structural distance $d=71$, QAP obj=49]: the master played Rxc4; under μ our *Re1* plays the role of that game’s *Rc1*; cross-type roles: our *Ra8* as their *Ph2*, our *Re8* as their *Pb3*; transferred move e8e7. (c) Retrieved from `lichess_elite_2024-09_p0016#857` (ply 35) [structural distance $d=59$, QAP obj=45]: the master played Kf7; under μ our *Kg1* plays the role of that game’s *Kg1*; cross-type roles: our *Qg4* as their *Nb8*, our *be7* as their *Rf8*; transferred move g8g7. (d) Retrieved from `lichess_elite_2024-09_p0016#1109` (ply 42) [structural distance $d=71$, QAP obj=45]: the master played Qh4; under μ our *Qg5* plays the role of that game’s *Qg5*; cross-type roles: our *Pc7* as their *Rc1*, our *Pa7* as their *Re1*; transferred move g4g5. Synthesis: no retrieved analog transferred to cxd6 – the `search` selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



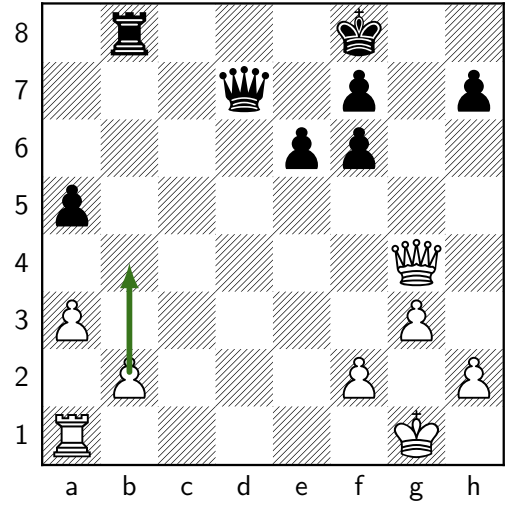
(a)



(b)

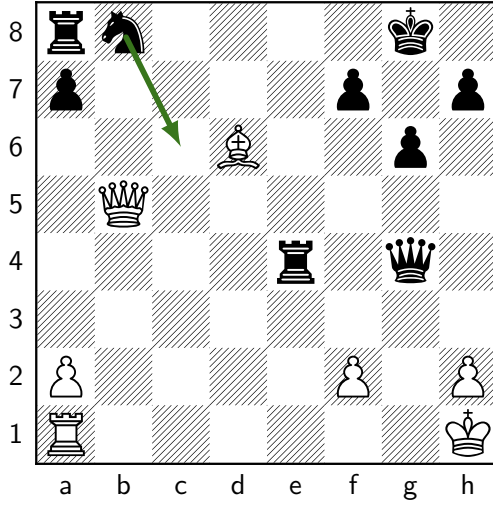


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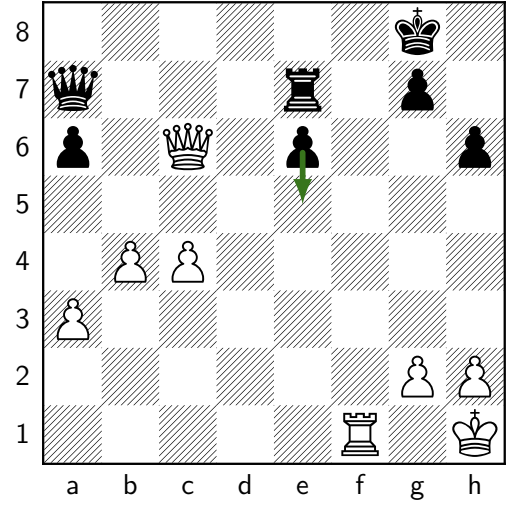


(d)

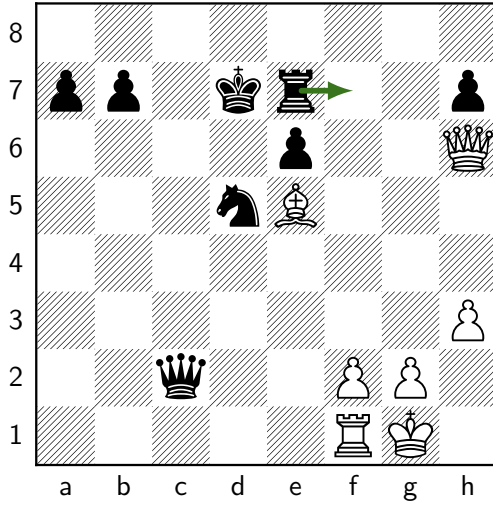
Figure 14: **Move 20, Black: Rxe4.** (a) Our position; Melanie plays **Rxe4** (e8e4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0031#3831` (ply 53) [structural distance $d=57$, QAP obj=36]: the master played **Qf8**; under μ our *Pg3* plays the role of that game’s *Qa6* (a **cross-type** correspondence); cross-type roles: our *Qg4* as their *Rd8*, our *bd6* as their *Pa7*; transferred move **g6g5**. (c) Retrieved from `lichess_elite_2024-09_p0022#1015` (ply 48) [structural distance $d=65$, QAP obj=35]: the master played **Nf3**; under μ our *Nb1* plays the role of that game’s *Nd4*; cross-type roles: our *Ra8* as their *Pb2*, our *bd6* as their *qe4*; transferred move **b8c6**. (d) Retrieved from `lichess_elite_2024-09_p0026#3274` (ply 52) [structural distance $d=64$, QAP obj=35]: the master played **b4**; under μ our *Pa2* plays the role of that game’s *Pb2*; cross-type roles: our *Nb8* as their *Pa3*, our *bd6* as their *pf7*; transferred move **a7a6**. Synthesis: no retrieved analog transferred to **Rxe4** – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



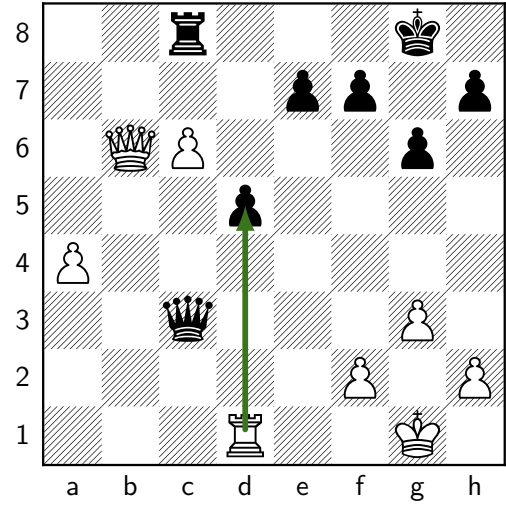
(a)



(b)

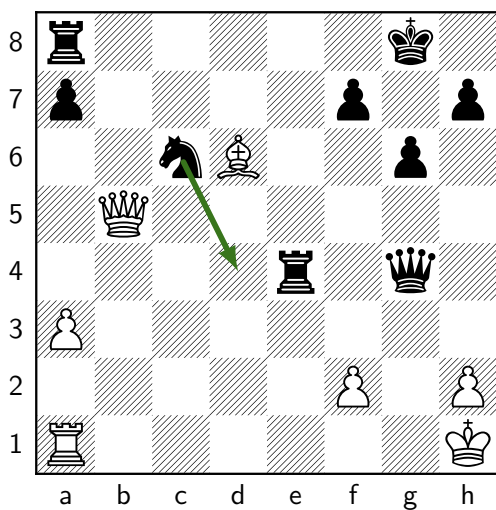


(c)

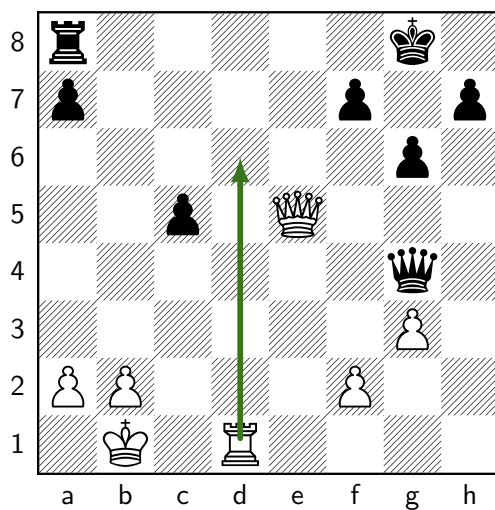


(d)

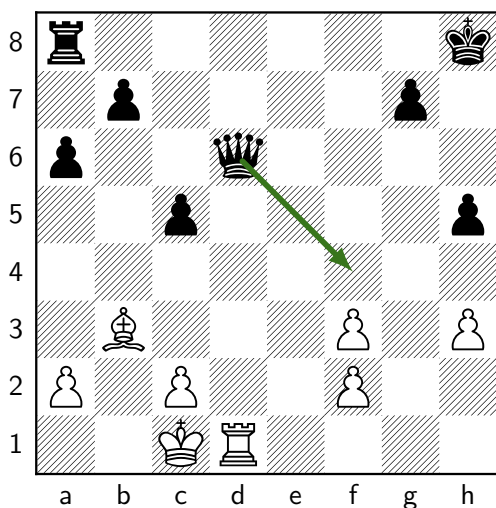
Figure 15: **Move 21, Black: Nc6**. (a) Our position; Melanie plays Nc6 (b8c6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0005#3875` (ply 55) [structural distance $d=63$, QAP obj=13]: the master played e5; under μ our Nb1 plays the role of that game’s Pe3 (a **cross-type** correspondence); cross-type roles: our Nb8 as their Pa6, our Re4 as their pb4; transferred move b8c6 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0019#1952` (ply 53) [structural distance $d=64$, QAP obj=15]: the master played Rf7; under μ our Qg5 plays the role of that game’s Re2 (a **cross-type** correspondence); cross-type roles: our Nb8 as their Ph7, our Nb8 as their Pe6; transferred move g4g5. (d) Retrieved from `lichess_elite_2024-09_p0021#2551` (ply 52) [structural distance $d=70$, QAP obj=13]: the master played Rxd5; under μ our Qg5 plays the role of that game’s Rd1 (a **cross-type** correspondence); cross-type roles: our ra1 as their Ph2, our Re4 as their Pg3; transferred move g4g5. Synthesis: 1 retrieved analog transferred to Nc6 (summed QAP objective 13, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



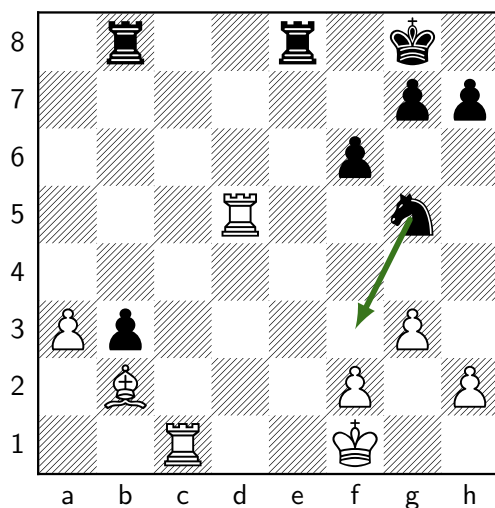
(a)



(b)

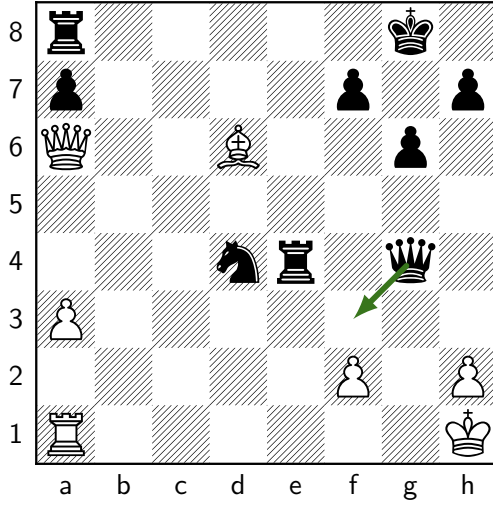


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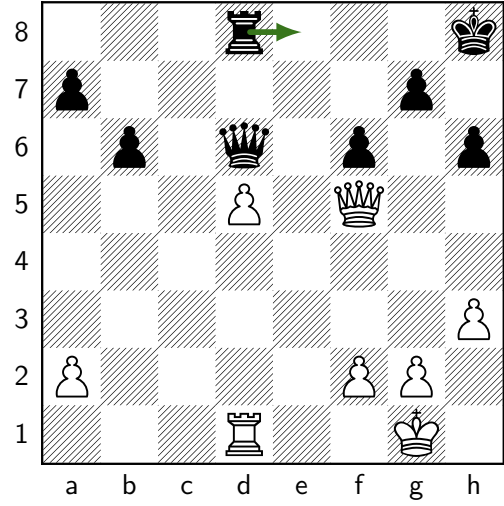


(d)

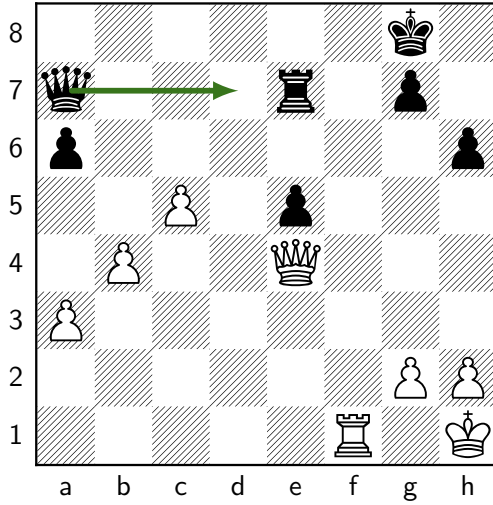
Figure 16: **Move 22, Black: Nd4**. (a) Our position; Melanie plays Nd4 (c6d4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0024#907` (ply 48) [structural distance $d=74$, QAP obj=15]: the master played Rd6; under μ our Nc3 plays the role of that game’s Rd1 (a **cross-type** correspondence); cross-type roles: our Nc6 as their Pg3, our bd6 as their pc5; transferred move c6d4 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0021#2982` (ply 55) [structural distance $d=71$, QAP obj=13]: the master played Qf4+; under μ our Nc3 plays the role of that game’s Qd3 (a **cross-type** correspondence); cross-type roles: our Pf7 as their Ra8, our ra1 as their Pg7; transferred move c6d4 (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0022#1954` (ply 59) [structural distance $d=71$, QAP obj=9]: the master played Nf3; under μ our Nc3 plays the role of that game’s Ng4; cross-type roles: our qb5 as their rd5, our Qg4 as their bb2; transferred move c6d4 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Nd4 (summed QAP objective 36, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



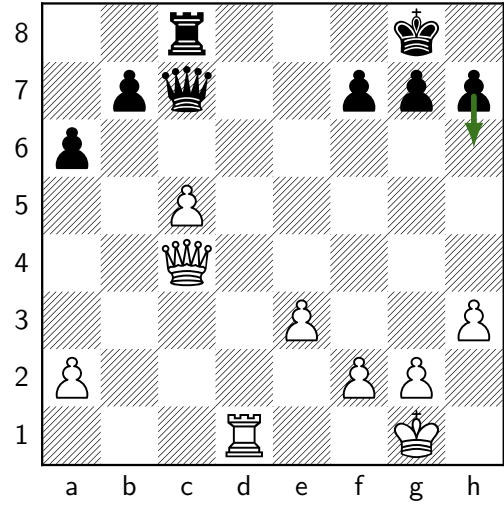
(a)



(b)

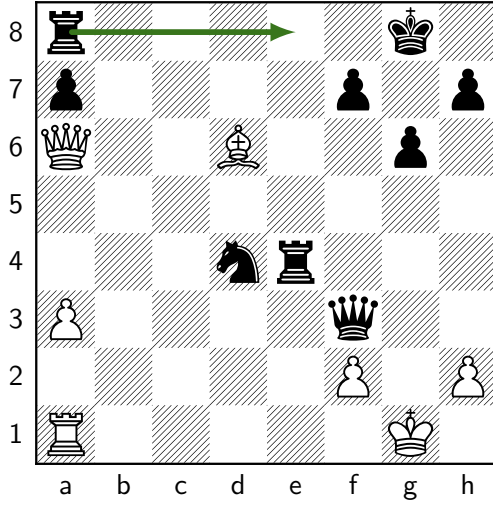


(c)

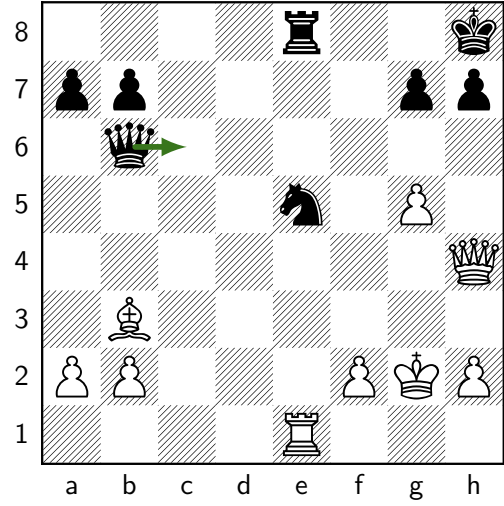


(d)

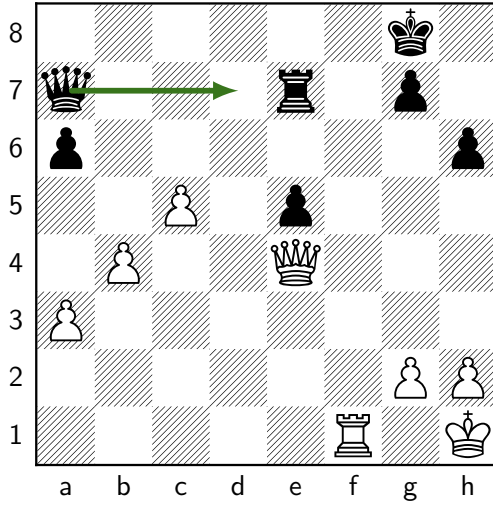
Figure 17: **Move 23, Black: Qf3+.** (a) Our position; Melanie plays Qf3+ (g4f3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0026#121` (ply 61) [structural distance $d=61$, QAP obj=52]: the master played Re8; under μ our Ra1 plays the role of that game’s Rd1; cross-type roles: our bd6 as their Pb6, our Nd4 as their Ph6; transferred move a8b8. (c) Retrieved from `lichess_elite_2024-09_p0005#3875` (ply 61) [structural distance $d=54$, QAP obj=46]: the master played Qd7; under μ our Re5 plays the role of that game’s Qa2 (a **cross-type** correspondence); cross-type roles: our Nd4 as their Pg7, our bd6 as their pb4; transferred move e4e8. (d) Retrieved from `lichess_elite_2024-09_p0005#3177` (ply 53) [structural distance $d=64$, QAP obj=32]: the master played h6; under μ our Ph2 plays the role of that game’s Ph2; cross-type roles: our bd6 as their Rc8, our Ra8 as their Pb7; transferred move h7h5. Synthesis: no retrieved analog transferred to Qf3+ – the search selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



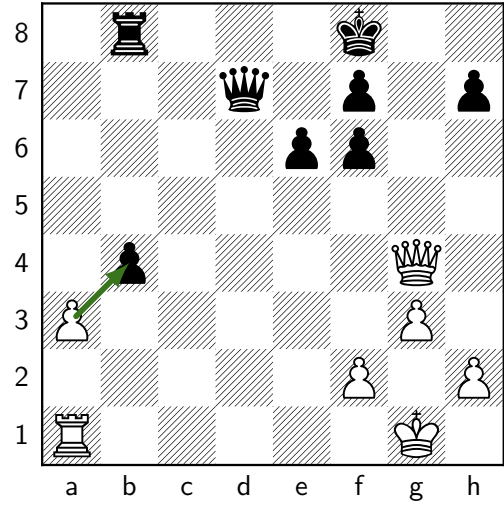
(a)



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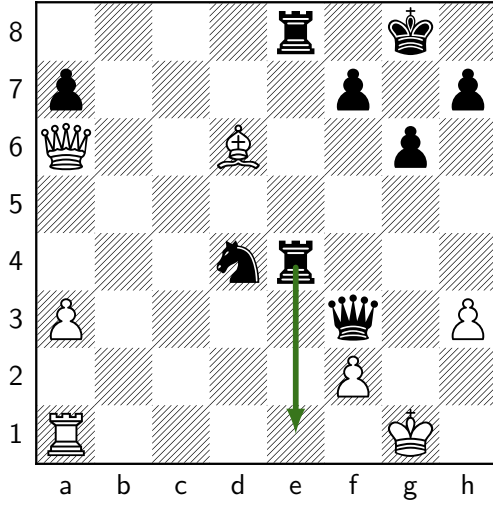


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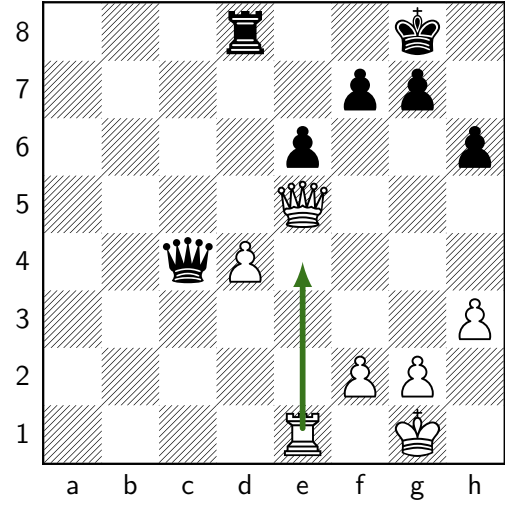


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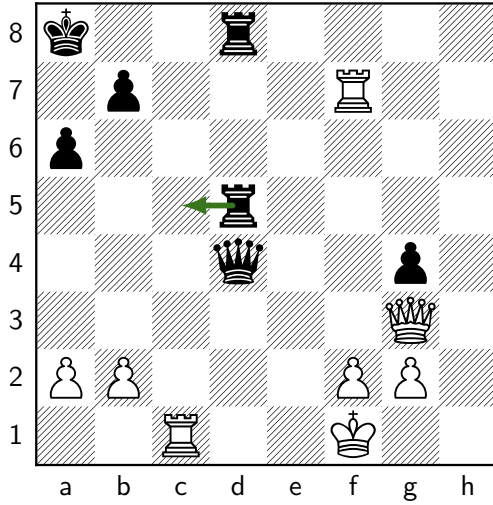
Figure 18: **Move 24, Black: Rae8.** (a) Our position; Melanie plays **Rae8** (a8e8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0018#772` (ply 59) [structural distance $d=57$, QAP obj=56]: the master played **Qc6+**; under μ our *Qf6* plays the role of that game’s *Qb3*; cross-type roles: our *Re4* as their *Pa7*, our *Nd4* as their *Pb7*; transferred move **f3g4**. (c) Retrieved from `lichess_elite_2024-09_p0005#3875` (ply 61) [structural distance $d=55$, QAP obj=56]: the master played **Qd7**; under μ our *Qf6* plays the role of that game’s *Qa2*; cross-type roles: our *Nd4* as their *Pg7*, our *Re4* as their *Pa6*; transferred move **f3g4**. (d) Retrieved from `lichess_elite_2024-09_p0026#3274` (ply 54) [structural distance $d=53$, QAP obj=43]: the master played **axb4**; under μ our *Qf6* plays the role of that game’s *Pa3* (a **cross-type** correspondence); cross-type roles: our *bd6* as their *Ra1*, our *Re4* as their *pf6*; transferred move **f3h5**. Synthesis: no retrieved analog transferred to **Rae8** – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



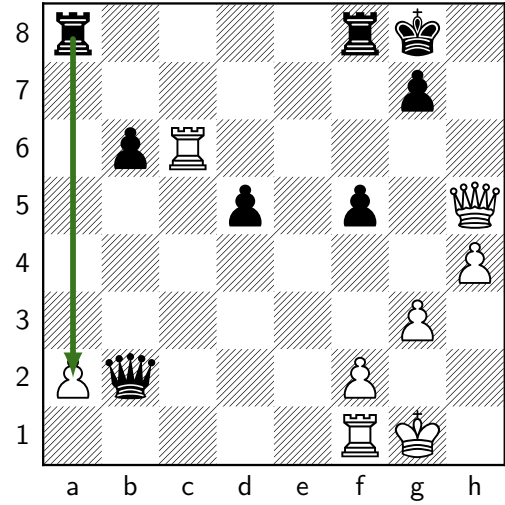
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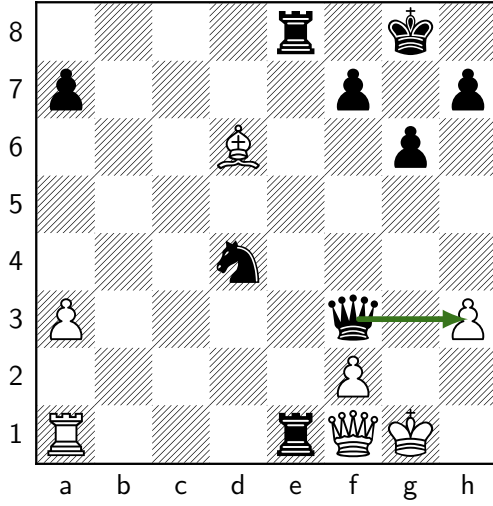


(c)

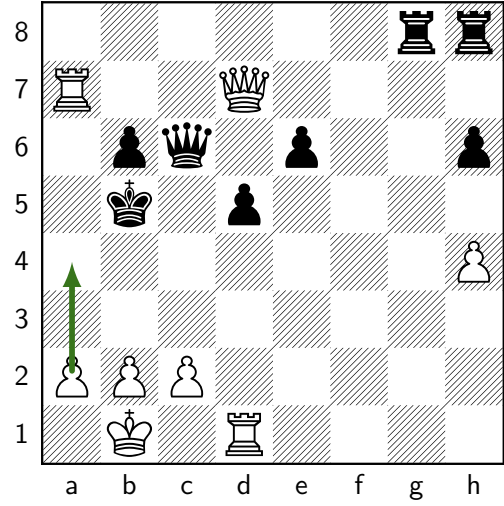


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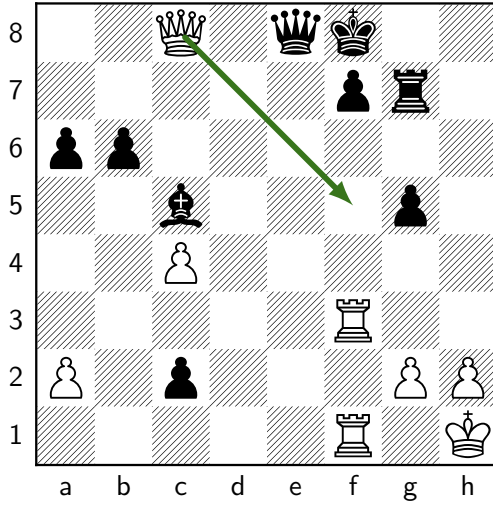
Figure 19: **Move 25, Black: Re1+.** (a) Our position; Melanie plays Re1+ (e4e1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0023#1636` (ply 56) [structural distance $d=62$, QAP obj=59]: the master played Re4; under μ our Re1 plays the role of that game’s Re1; cross-type roles: our Re4 as their ph6, our bd6 as their pf7; transferred move e8e5. (c) Retrieved from `lichess_elite_2024-09_p0022#2913` (ply 45) [structural distance $d=54$, QAP obj=48]: the master played Rc5; under μ our Re5 plays the role of that game’s Rd4; cross-type roles: our bd6 as their rf7; transferred move e4e5. (d) Retrieved from `lichess_elite_2024-09_p0018#1996` (ply 47) [structural distance $d=54$, QAP obj=45]: the master played Rxa2; under μ our Pg3 plays the role of that game’s Ra1 (a **cross-type** correspondence); cross-type roles: our qa6 as their rc6, our Nd4 as their ph4; transferred move g6g5. Synthesis: no retrieved analog transferred to Re1+ – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



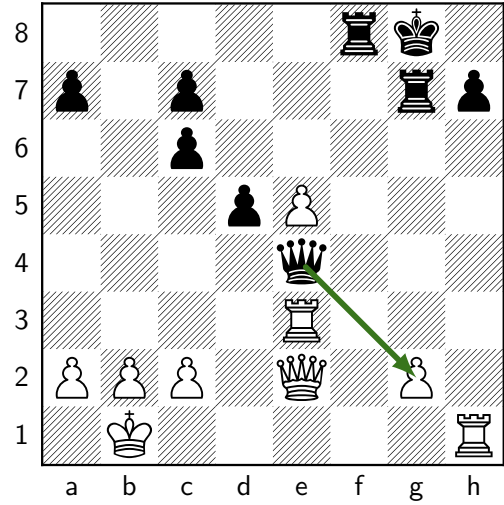
(a)



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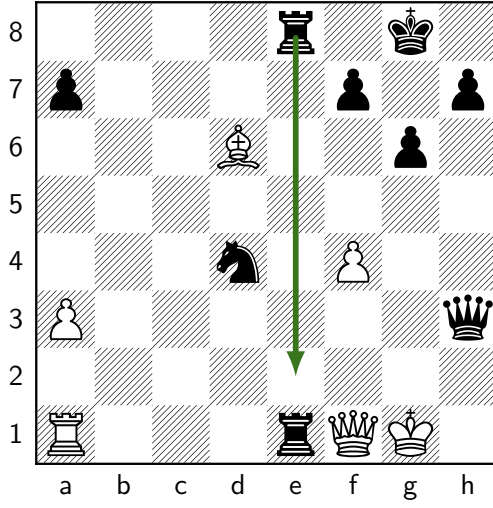


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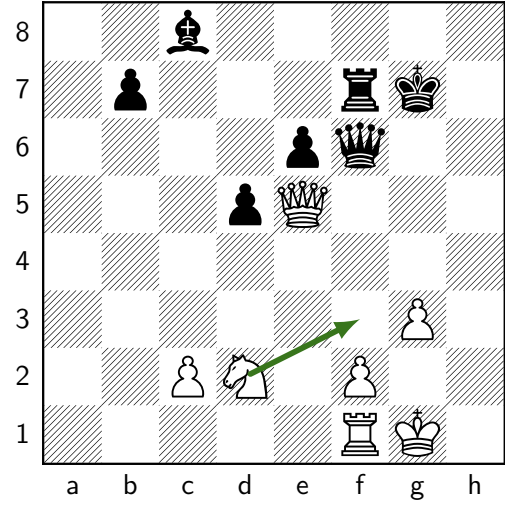


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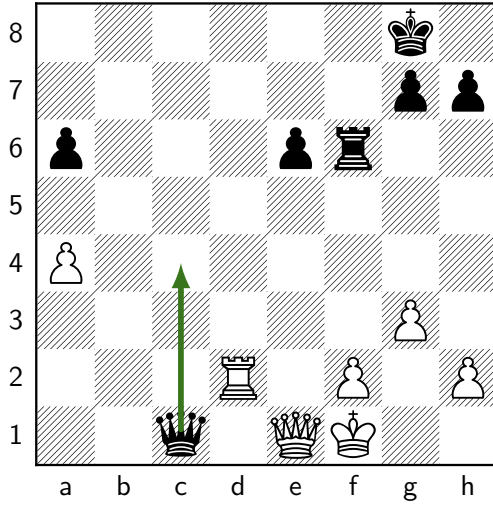
Figure 20: **Move 26, Black: Qxh3.** (a) Our position; Melanie plays Qxh3 (f3h3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0013#1586` (ply 66) [structural distance $d=61$, QAP obj=61]: the master played a4+; under μ our Pf2 plays the role of that game’s Pa2; cross-type roles: our bd6 as their Pc2, our Nd4 as their pd5; transferred move f7f6. (c) Retrieved from `lichess_elite_2024-09_p0031#1103` (ply 56) [structural distance $d=62$, QAP obj=42]: the master played Qf5; under μ our Re8 plays the role of that game’s Qc8 (a **cross-type** correspondence); cross-type roles: our bd6 as their pc2, our Nd4 as their Pg2; transferred move e1e4. (d) Retrieved from `lichess_elite_2024-09_p0024#732` (ply 61) [structural distance $d=72$, QAP obj=36]: the master played Qxg2; under μ our Re8 plays the role of that game’s Qe5 (a **cross-type** correspondence); cross-type roles: our Nd4 as their Pc7, our Ph7 as their Rg7; transferred move e1d1. Synthesis: no retrieved analog transferred to Qxh3 – the search selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



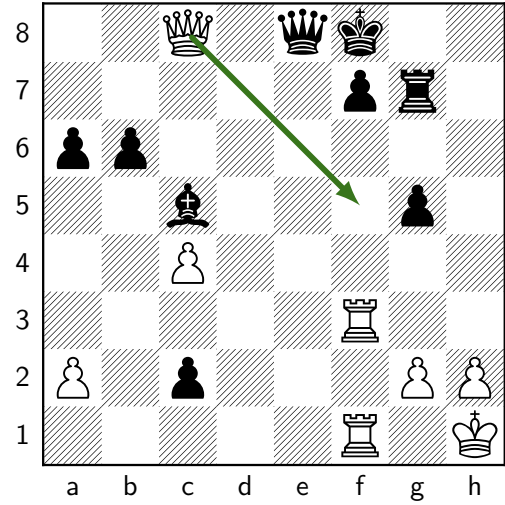
(a)



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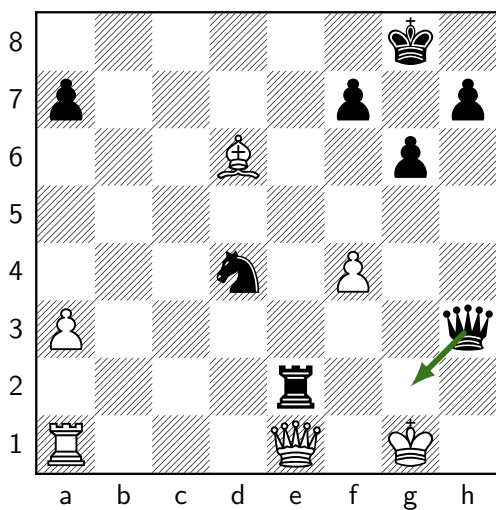


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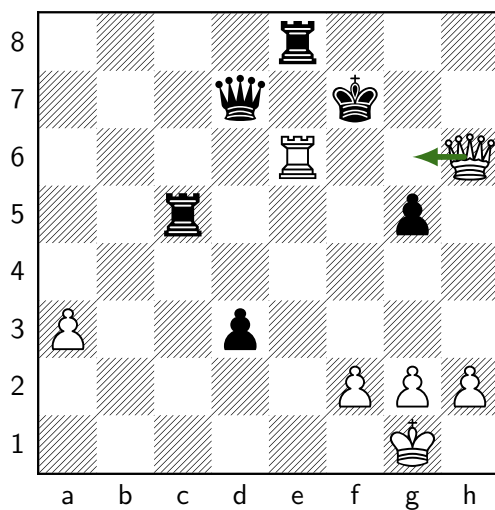


(d)

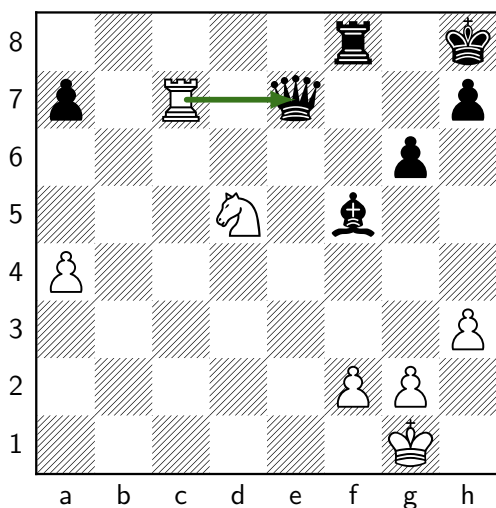
Figure 21: **Move 27, Black: R8e2**. (a) Our position; Melanie plays R8e2 (e8e2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0010#2715` (ply 48) [structural distance $d=69$, QAP obj=59]: the master played Nf3; under μ our Pg3 plays the role of that game’s Nd2 (a **cross-type** correspondence); cross-type roles: our Qh3 as their Pf2, our Re1 as their Qe5; transferred move g6g5. (c) Retrieved from `lichess_elite_2024-09_p0031#3258` (ply 61) [structural distance $d=75$, QAP obj=45]: the master played Qc4+; under μ our Re8 plays the role of that game’s Qc8 (a **cross-type** correspondence); cross-type roles: our ra1 as their Pg7, our Nd4 as their Pe6; transferred move e1e4. (d) Retrieved from `lichess_elite_2024-09_p0031#1103` (ply 56) [structural distance $d=70$, QAP obj=39]: the master played Qf5; under μ our Re8 plays the role of that game’s Qc8 (a **cross-type** correspondence); cross-type roles: our bd6 as their pc2, our Nd4 as their Ph2; transferred move e1e4. Synthesis: no retrieved analog transferred to R8e2 – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



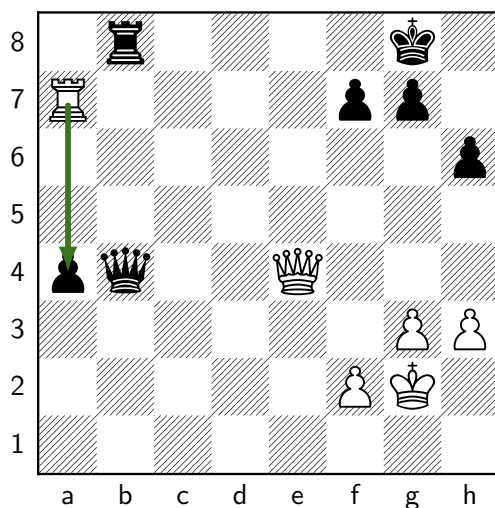
(a)



(b)



(c)



(d)

Figure 22: **Move 28, Black: Qg2#.** (a) Our position; Melanie plays Qg2# (h3g2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0024#1601` (ply 70) [structural distance $d=54$, QAP obj=22]: the master played Qg6+; under μ our Qh6 plays the role of that game’s Qh6; cross-type roles: our bd6 as their Ph2, our ra1 as their pd3; transferred move h3h2. (c) Retrieved from `lichess_elite_2024-09_p0024#1488` (ply 58) [structural distance $d=48$, QAP obj=21]: the master played Rxe7; under μ our Re7 plays the role of that game’s Rc7; cross-type roles: our bd6 as their Pg2, our Qh3 as their Ph3; transferred move e2c2. (d) Retrieved from `lichess_elite_2024-09_p0013#3967` (ply 56) [structural distance $d=44$, QAP obj=15]: the master played Rxa4; under μ our Qh6 plays the role of that game’s Ra7 (a **cross-type** correspondence); cross-type roles: our bd6 as their pa4, our Re2 as their Qe4; transferred move h3h4. Synthesis: no retrieved analog transferred to Qg2# – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.